

Agent-based Modeling: Equilibrium, Echo Chambers, and Efficiency in Hybrid Coevolutionary Opinion Games

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Abstract

The rise of echo chambers and polarization in online social networks poses fundamental questions about the efficiency and stability of collective opinion formation. Traditional numerical opinion models provide formal tools for analyzing social cost and equilibrium, but often simplify communication into predefined update rules. Meanwhile, LLM-driven agents capture richer language-based reasoning and memory, yet neither paradigm accounts for their joint social cost.

We close this gap with the Hybrid Coevolutionary Opinion Game (H-COG) framework, integrating cost-minimizing Friedkin–Johnsen agents and Phi-4 LLM-driven reasoning agents in a dynamically rewired social network. Agent beliefs are initialized from 5,199 real-world Reddit stance comments, scored on a continuous $[-1, +1]$ scale via a fine-tuned RoBERTa regressor, and a mixing parameter α controls the agent composition.

Our first result is that the coevolving graph converges under LLM-driven updating as reliably as under an optimizing update rule with a known fixed point, which is what licenses reading a Price of Anarchy from it at all. On that basis, purely LLM-driven populations are roughly five times less efficient than purely analytical ones, and a decomposition of social cost locates the gap: language agents are displaced from their own prior positions rather than merely disagreeing with their neighbors. The same susceptibility makes them less polarized, so lower polarization here is a symptom of what makes them inefficient rather than evidence of better deliberation. The pattern holds across three initial topologies.

Keywords: Opinion dynamics, Echo chambers, Price of Anarchy, Language models, Agent-based modeling

1 Introduction

Online platforms have turned the public square into a coevolving system: what a user reads shapes what they think, and what they think reshapes whom they follow. Echo chambers are the visible result. A population that mostly meets what already agrees with it polarizes further and carries false claims farther, which is why the phenomenon has become a societal concern. Explaining it means tracking opinions and ties together, not either alone.

Two lines of work approach this, and each gives up something the other keeps. Analytical opinion models supply explicit update rules and formal tools for equilibrium, social cost and the Price of Anarchy (PoA), among them DeGroot, Friedkin–Johnsen, bounded confidence and the coevolutionary opinion games [DeGroot, 1974, Friedkin and Johnsen, 1990, Bhawalkar et al., 2013]. What they give up is language: each

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exchange becomes a numerical update, and the semantic content through which people actually persuade one another drops out. LLM-based agent simulations keep that content, letting agents read, reason and remember, and they reproduce echo chambers and polarization [Wang et al., 2025, Gu et al., 2025]. What they give up is the yardstick. These studies aim at behavioral realism in populations of one kind, so no framework yet measures social cost and PoA where language-driven and cost-driven agents share a network.

H-COG closes that gap by putting both kinds of agent in one network that rewires as opinions change. A Type-C agent minimizes a numerical disagreement cost, while a Type-L agent reads what its neighbors wrote and answers in language. The parameter α fixes how many of each the population holds, both are read on the same continuous scale, and every agent re-chooses its neighbors by closeness of position. Sweeping α therefore compares purely numerical, purely LLM-driven and mixed populations under one set of conditions. Efficiency is measured by social cost and a topology-conditioned PoA, with the observed cost and the planner optimum computed on the same realized network after each rewiring.

The simulations rest on 5,199 `r/politics` comments about gun control and abortion. A fine-tuned RoBERTa regressor maps each comment to a continuous stance in $[-1, +1]$, and stratified samples from the resulting distributions set the agents’ intrinsic opinions. The two topics differ in how their stances are spread, which lets us ask whether the effect of composition survives a change in the starting landscape. We sweep nine mixing levels, $\alpha \in \{0, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875, 1\}$, across three initial topologies.

1.1 Research Questions

The study addresses the following questions:

- RQ1.** How does the proportion of Type-C and Type-L agents affect social cost and the Price of Anarchy?
- RQ2.** How does agent composition affect polarization, echo-chamber structure, and network dynamics?
- RQ3.** Which component of social cost accounts for the efficiency difference between LLM-driven and cost-minimizing populations?

1.2 Contributions

This study makes four contributions:

1. **Hybrid coevolutionary framework.** We introduce H-COG, which integrates cost-minimizing Type-C agents and LLM-driven Type-L agents in a common network whose opinions and connections coevolve.
2. **Empirically grounded initialization.** We construct a two-stage pipeline that extracts real-world stance distributions from Reddit using semantic clustering and a fine-tuned RoBERTa regressor, then initializes agents through stratified sampling.
3. **Common efficiency evaluation.** We formulate social cost and topology-conditioned PoA on a shared opinion scale, and decompose social cost into disagreement and conformity components to identify the source of efficiency differences.
4. **Systematic population comparison.** We compare nine Type-C/Type-L mixing levels across two topics and three initial topologies, evaluating efficiency, polarization, echo-chamber structure, and convergence behavior.

Section 2 reviews the two research traditions on which H-COG builds. Section 3 formalizes the model and its coevolutionary dynamics. Section 4 describes the data and simulation methodology, Section 5 presents the equilibrium analysis, and Section 6 reports the experimental design and numerical results. Section 7 discusses the findings, limitations, and directions for future work.

2 Related Work

2.1 Opinion Dynamics and Coevolutionary Games

Opinion dynamics begins with agents that hold a number and update it by averaging their neighbors. DeGroot formalized that process and showed that under mild conditions repeated averaging drives the whole network to consensus DeGroot [1974]. Friedkin and Johnsen refined it by separating the belief an agent holds privately from the one it expresses, which lets the system settle where disagreement survives instead of vanishing Friedkin and Johnsen [1990].

Bhawalkar et al. took the further step of letting the network move as well Bhawalkar et al. [2013]. In their coevolutionary opinion formation game each agent attaches to neighbors who think as it does and picks an expressed opinion that trades social pressure against private belief. That setup admits a formal Price of Anarchy (PoA) analysis: how far does a decentralized outcome fall short of what a central planner could reach? Chen et al. carried it to a wider class of directed graphs Chen et al. [2016]. The loss stays bounded so long as no single node is too influential. All of these models speak in numerical update rules. What they leave out is the reasoning, the memory and the meaning through which people actually form and state an opinion.

2.2 LLM-based Agent Simulation for Echo Chambers

Large language models opened a different route. Instead of following a fixed numerical rule, an agent can read text, reason about it and answer in kind, which makes its behavior far more expressive and far more sensitive to context. Park et al. showed that such agents produce believable social behavior in interactive settings, which laid the ground for using them at network scale Park et al. [2023].

The idea reached echo chambers quickly. Wang et al. built a language-based framework in which agents talk, revise what they believe by reasoning, and rewire on closeness of opinion Wang et al. [2025]. They found that these agents track observed polarization more closely than Friedkin–Johnsen or bounded-confidence models do. Gu et al. took a comparable route but grounded the simulation in real Twitter data, which let them check the framework against observed behavior and compare how well different LLMs capture echo-chamber dynamics Gu et al. [2025].

Together these studies show that language-based agents reach something analytical models cannot. Both, though, run a single kind of agent throughout, and both report descriptive indices such as polarization and modularity. Neither asks what the resulting equilibria cost. Whether reasoning in language leaves a population better or worse off is therefore still open.

3 Model and Problem Formulation

3.1 Agents and Opinions

Let $\mathcal{N} = \{1, \dots, N\}$ be a set of N agents. Each agent i has an **intrinsic opinion** $s_i \in [-1, +1]$, representing a fixed prior stance, and an **expressed opinion** $z_i^{(t)} \in [-1, +1]$, which evolves as the agent responds to its neighbors.

The mixing parameter $\alpha \in [0, 1]$ partitions the population into two types. **Type-C** agents (fraction α) update their expressed opinions by minimizing a local disagreement cost derived from the Friedkin–Johnsen model [Friedkin and Johnsen, 1990]. **Type-L** agents (fraction $1 - \alpha$) generate a natural-language opinion $o_i^{(t)} \in \mathcal{V}^*$, where $\mathcal{V}$ is a finite token vocabulary. A continuous stance mapping

$$e : \mathcal{V}^* \rightarrow [-1, +1]$$

places that opinion on the common numerical scale: $z_i^{(t)} = e(o_i^{(t)})$. Type-C agents directly produce numerical opinions on the same scale. The cases $\alpha = 1$ and $\alpha = 0$ give the purely numerical and purely LLM-driven baselines, respectively.

3.2 State Space and Opinion Updates

The global state at step t is

$$S^{(t)} = \left(G^{(t)}, \mathbf{X}^{(t-1)} \right),$$

where $G^{(t)} = (S_1^{(t)}, \dots, S_N^{(t)})$ is the directed network, $S_i^{(t)} \subset \mathcal{N} \setminus \{i\}$ is agent i 's out-neighborhood with $|S_i^{(t)}| = K$, and $\mathbf{X}^{(t-1)} = (\mathbf{x}_1^{(t-1)}, \dots, \mathbf{x}_N^{(t-1)})$ collects the preceding structured text records of the Type-L agents.

Type-C analytical agents

A Type-C agent chooses an expressed opinion from $\mathcal{A}_i = [-1, +1]$. Holding its neighbors' latest opinions fixed, its myopic stage-cost minimizer is

$$z_i^{(t+1)} = \arg \min_{z \in [-1, +1]} C_i \left(G^{(t)}, z, \mathbf{z}_{-i}^{(t)} \right) = \frac{\sum_{j \in S_i^{(t)}} z_j^{(t)} + \rho_i K s_i}{K(1 + \rho_i)}. \quad (1)$$

Because this expression is a convex combination of values in $[-1, +1]$, the opinion domain is invariant. In the Markov-game representation, a more general Type-C policy $\pi_i^C(\cdot \mid S^{(t)})$ may instead optimize expected discounted costs; the simulations use the myopic update in Eq. (1).

Type-L generative agents

A Type-L action is a structured sequence $\mathbf{x}_i^{(t+1)} \in \mathcal{V}^*$ containing reasoning, opinion, and memory in that order. Its prompt combines the previous record, current neighbor messages, and intrinsic stance:

$$P_i^{(t)} = \text{Prompt} \left(\mathbf{x}_i^{(t)}, \{\mathbf{x}_j^{(t)}\}_{j \in S_i^{(t)}}, s_i \right).$$

An autoregressive LLM with parameters θ generates $\mathbf{x}_i^{(t+1)} = (y_1, \dots, y_m)$ according to

$$\mathbb{P}_\theta \left(\mathbf{x}_i^{(t+1)} \mid P_i^{(t)} \right) = \prod_{k=1}^m \mathbb{P}_\theta \left(y_k \mid y_{<k}, P_i^{(t)} \right).$$

The simulations use deterministic decoding. Each agent makes one LLM call per step; the memory field is rewritten and limited to 120 words. Only the free-form opinion field $o_i^{(t+1)}$ is passed to the stance mapping, giving $z_i^{(t+1)} = e(o_i^{(t+1)}) \in [-1, +1]$.

3.3 Coevolutionary Rewiring

After all opinions are updated, the network changes from $G^{(t)}$ to $G^{(t+1)}$. Let $d_{ij}^{(t+1)} = |z_j^{(t+1)} - s_i|$ be the distance between peer j 's expressed opinion and agent i 's intrinsic opinion. A general stochastic rewiring kernel assigns attachment weight

$$w_{ij}^{(t+1)} = \exp \left(-\beta d_{ij}^{(t+1)} \right), \quad \beta > 0,$$

and draws each K -subset S with probability

$$\mathbb{P}\left(S_i^{(t+1)} = S \mid \mathbf{z}^{(t+1)}, s_i\right) = \frac{\prod_{j \in S} w_{ij}^{(t+1)}}{\sum_{\substack{S' \subset \mathcal{N} \setminus \{i\} \\ |S'|=K}} \prod_{k \in S'} w_{ik}^{(t+1)}}.$$

The experiments use the $\beta \rightarrow \infty$ limit:

$$S_i^{(t+1)} = \arg \operatorname{top-K}_{j \neq i} \left(-|s_i - z_j^{(t+1)}| \right), \quad K = 5. \quad (2)$$

Each agent therefore connects to the five peers whose expressed opinions lie nearest its own intrinsic stance, with ties broken by a seeded random draw. Rewiring is a kernel of the environment, not an action any agent chooses.

3.4 Social Cost and Price of Anarchy

Regardless of agent type, agent i incurs the stage cost

$$C_i\left(G^{(t)}, \mathbf{z}^{(t)}\right) = \sum_{j \in S_i^{(t)}} (z_i^{(t)} - z_j^{(t)})^2 + \rho_i K (z_i^{(t)} - s_i)^2.$$

The global social cost is the sum of individual costs:

$$SC(G, \mathbf{z}) = \underbrace{\sum_i \sum_{j \in S_i} (z_i - z_j)^2}_{\text{disagreement}} + \underbrace{\sum_i \rho_i K (z_i - s_i)^2}_{\text{conformity}}. \quad (3)$$

The first component measures disagreement with current neighbors, while the second measures displacement from intrinsic opinions.

Definition 1 (Nash Equilibrium). *A joint strategy profile $\pi^* = (\pi_1^*, \dots, \pi_N^*)$ is a Nash equilibrium if:*

1. *for every Type-C agent $i \in \mathcal{N}_C$, π_i^* minimizes its expected discounted cumulative stage cost given π_{-i}^* ,*

$$\pi_i^* \in \arg \min_{\pi_i} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t C_i(G^{(t)}, \mathbf{z}^{(t)}) \mid S^{(0)} \right];$$

2. *for every Type-L agent $j \in \mathcal{N}_L$, π_j^* is induced by the autoregressive LLM kernel $\mathbb{P}_{\theta}(\mathbf{x} \mid P_j)$.*

Definition 2 (Generalized Price of Anarchy). *Let $\Omega_{\text{NE}}(\alpha)$ denote the stationary state-action distributions resulting from an equilibrium at mixture level α . The generalized hybrid PoA is*

$$\text{PoA}(\alpha) = \frac{\max_{\mu \in \Omega_{\text{NE}}(\alpha)} \mathbb{E}_{(G, \mathbf{z}) \sim \mu} [SC(G, \mathbf{z})]}{\min_{G^*, \mathbf{z}^*} SC(G^*, \mathbf{z}^*)}. \quad (4)$$

The experiments report a topology-conditioned operational measure. At step t , the planner optimizes opinions on the same realized graph:

$$\mathbf{z}_t^* = \arg \min_{\mathbf{z} \in [-1, +1]^N} SC(G^{(t)}, \mathbf{z}),$$

and

$$\text{PoA}^{(t)} = \frac{SC(G^{(t)}, \mathbf{z}^{(t)})}{SC(G^{(t)}, \mathbf{z}_t^*)} \geq 1. \quad (5)$$

Fixing the graph isolates the welfare loss associated with decentralized opinion updating from that associated with the topology produced by rewiring. We report both components of Eq. (3) and their ratio.

3.5 Evaluation Metrics

Polarization (P_z). Following Chitra and Musco [2020], polarization is the variance of expressed opinions:

$$P_z^{(t)} = \frac{1}{N} \sum_{i=1}^N \left(z_i^{(t)} - \bar{z}^{(t)} \right)^2, \quad \bar{z}^{(t)} = \frac{1}{N} \sum_i z_i^{(t)}. \quad (6)$$

Higher P_z indicates greater opinion divergence.

Modularity (Q and Q_{norm}). Louvain community detection [Blondel et al., 2008] is applied to the undirected projection of $G^{(t)}$, yielding modularity $Q_{\text{obs}}(t)$ [Newman, 2004]. Because random graphs can exhibit nonzero modularity from degree fluctuations alone [Guimerà et al., 2004], we compare the observed graph with a degree-preserving null model:

$$Q_{\text{norm}}(t) = Q_{\text{obs}}(t) - \bar{Q}_{\text{rand}}(t), \quad z_Q(t) = \frac{Q_{\text{obs}}(t) - \bar{Q}_{\text{rand}}(t)}{\sigma_{\text{rand}}(t)}. \quad (7)$$

Here Q_{norm} measures modularity attributable to opinion-based sorting relative to the same degree sequence, and z_Q expresses the difference in units of the null distribution’s standard deviation.

Structural convergence. The consecutive-graph measures $C^{\text{out}}(t)$, $dS(t)$, $d_L(t)$, and DeltaCon(t) are defined in Section 4.4.2 and logged at every step transition.

4 Methodology

4.1 Simulation Framework Overview

Figure 1 summarizes the two-stage pipeline. **Stage 1** converts raw Reddit comments into calibrated stance distributions for agent initialization. Comments are semantically clustered, scored with a fine-tuned RoBERTa regressor, and sampled by stance stratum to produce $N = 50$ agents whose intrinsic-opinion distribution follows the empirical population. **Stage 2** runs H-COG: the sampled agents are assigned to Type-C and Type-L according to α , update their expressed opinions, and coevolve with the network until the stopping rule in Section 4.4.3 is met. At each step we record social cost and PoA, polarization, null-corrected modularity, and the structural convergence measures in Section 4.4.2.

4.2 Stage 1: Reddit Data Pipeline

Corpus Processing and Topic Clustering

We source comments from the `r/politics` subreddit using `.zst`-compressed PushShift dumps. We retain non-deleted, non-removed top-level comments and, for each author and topic, keep only the highest-scoring comment to avoid within-author duplication. Comment bodies are encoded with the pre-trained Sentence-BERT model `all-MiniLM-L6-v2` [Reimers and Gurevych, 2019], cosine normalized, reduced

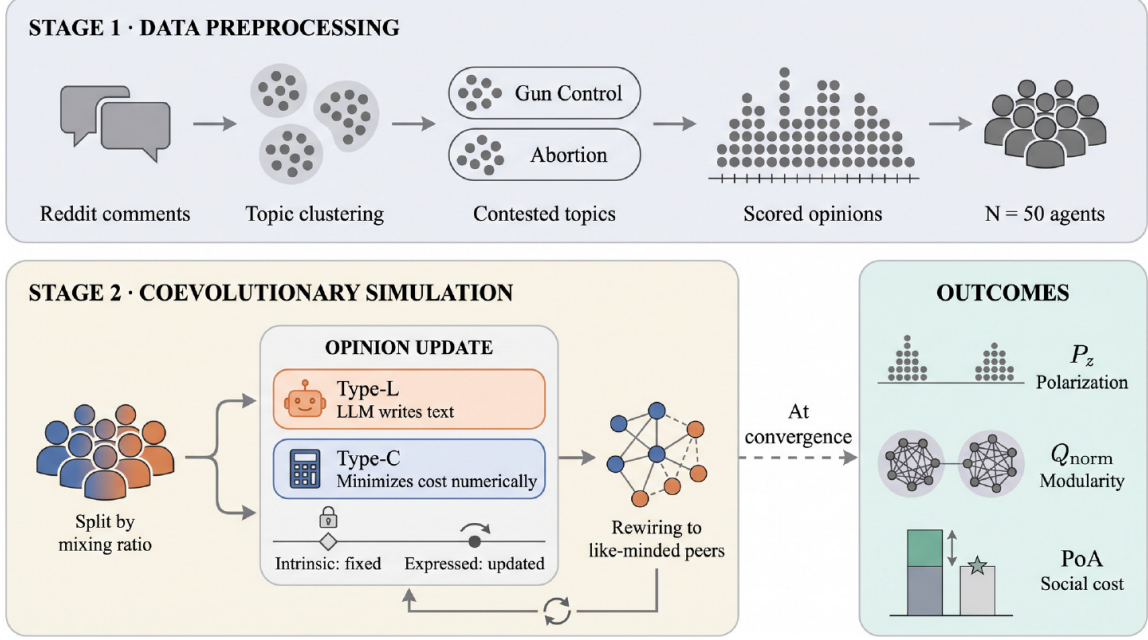


Figure 1: Two-stage pipeline: Stage 1 extracts stance distributions from Reddit; Stage 2 runs H-COG with heterogeneous agents on a coevolving network.

from 384 to 50 dimensions by UMAP with a cosine metric [McInnes et al., 2018], and clustered with HDBSCAN [Campello et al., 2013, McInnes et al., 2017]. We use `min_cluster_size=50`, `min_samples=5`, and EOM cluster selection, obtaining 1,184 topic clusters. Each cluster is labeled by the eight highest-weight terms in a cluster-level TF-IDF matrix.

Continuous Stance Measurement

Architecture. We replace the classification head of `RobertaForSequenceClassification` [Liu et al., 2019] with a regression head,

$$\hat{y}_i = \tanh(\mathbf{W}\mathbf{h}_{[\text{CLS}]}) \in [-1, +1], \quad (8)$$

where $\mathbf{h}_{[\text{CLS}]} \in \mathbb{R}^{768}$ is the pooled token representation and $\mathbf{W} \in \mathbb{R}^{1 \times 768}$. Negative and positive endpoints represent strong opposition and support, respectively.

Training. The regressor is trained on the IBM Argument Quality 30K corpus [Gretz et al., 2020], which contains crowd-sourced annotations for contested topics including gun control and abortion. Graded supervision is recovered through Best–Worst Scaling [Kiritchenko and Mohammad, 2017]. On held-out human judgments, the fitted scorer is monotone in group means and correlates with the graded target at $r = 0.926$. The same instrument scores the Reddit corpus, initializes intrinsic opinions, and measures Type-L expressed opinions, placing all three on the common scale defined in Section 3.1.

Stratified Agent Sampling

For each target topic, we select $N = 50$ agents as follows:

1. Score all comments to obtain the empirical stance distribution $\hat{P}(\text{stance})$.
2. Partition $[-1, +1]$ into five strata using thresholds $\{-0.75, -0.25, +0.25, +0.75\}$.

3. For each stratum b , compute its empirical proportion π_b and sample $n_b = \lfloor 50\pi_b \rfloor$ comments, adjusting rounding so that $\sum_b n_b = 50$.

The strata set sampling quotas and nothing else: agent i keeps the continuous score of its own source comment as s_i , not the stratum index or its midpoint.

4.3 Stage 2: H-COG Simulation

4.3.1 Agent Initialization

Each Type-C agent receives an intrinsic opinion s_i , a stubbornness $\rho_i \sim \text{Uniform}(0.3, 0.7)$ drawn independently of stance, and starts at $z_i^{(0)} = s_i$. A Type-L agent also carries a persona, written once by Phi-4 [Abdin et al., 2024] and then held fixed, giving it a name, an age, a gender, an education level and five personality traits, which follows the persona conditioning of generative-agent architectures [Park et al., 2023]. The same model writes its opening text from that persona and the stance keyword of the stratum it was drawn from. The keyword is scaffolding for one call. What the agent then says is free-form, and it is scored as $z_i^{(0)} = e(o_i^{(0)})$, exactly as at every later step.

4.3.2 Network Initialization

We manipulate the initial topology using three graph families: Barabási–Albert with preferential attachment $m = 2$ [Barabási and Albert, 1999], Erdős–Rényi [Erdős and Rényi, 1959], and Watts–Strogatz [Watts and Strogatz, 1998]. Their generation parameters match mean degree while preserving differences in clustering; the values are reported in Section 6.2. Ten independently seeded graphs from each family are crossed with nine α levels and two topics, giving $9 \times 3 \times 10 \times 2 = 540$ runs.

K-NN rewiring rebuilds all edges after every update, so the initial topology affects the first update directly and later dynamics only through that trajectory. Across seeds, we vary the topology and Type-C/Type-L assignment while holding the sampled agents, intrinsic opinions, personas, and stubbornness values fixed within each topic. Confidence intervals therefore measure topology-and-assignment variability rather than variability from drawing a new sample of Reddit users.

Type-L agents use deterministic decoding (temperature = 0), and the vLLM server is initialized with a fixed seed.

4.3.3 Opinion Updates

Type-C agents use the Friedkin–Johnsen stage-cost minimizer in Eq. (1). Type-L agents make one LLM call per step and return the structured fields `reasoning`, `opinion`, and `memory` in that order.

Placing `reasoning` first conditions the subsequent opinion on the preceding deliberation [Wei et al., 2022]; fixing the order also avoids changes in response format across steps [Turpin et al., 2023]. The `opinion` field is the agent’s only position output and is scored externally by the regressor,

$$z_i^{(t+1)} = e(o_i^{(t+1)}) \in [-1, +1].$$

We do not request a numerical self-rating because language models tend to regress toward the center of an offered scale [Maier et al., 2025] and anchor on round values [Sato et al., 2026]. External scoring also evaluates both agent types on the same scale. The `memory` field is rewritten at each step and limited to 120 words, controlling prompt growth while retaining a bounded summary of earlier interactions; it does not itself guarantee stationarity. Reasoning records are retained, and current neighbor opinions are supplied verbatim.

4.3.4 K-NN Dynamic Rewiring

After all opinions are updated, Eq. (2) rebuilds every out-neighborhood by connecting each agent to the $K = 5$ peers whose expressed opinions are closest to its intrinsic opinion. We analyze convergence of the graph $G^{(t)}$ because the model’s interaction structure may continue to change even when the opinion vector is nearly stationary. Detecting a stable long-run structural regime lets us compare conditions at corresponding stages of their dynamics rather than at an arbitrary common time index.

4.4 Structural Convergence and Stopping Rule

4.4.1 Convergence Criterion

We define convergence operationally as entry into an attractor or a stable structural regime: the rate of graph change ceases to decrease, although individual edges need not stop changing. K-NN reconstructs each out-neighborhood from a continuous ranking, so a small perturbation near the cutoff can replace an edge. Adaptive networks may consequently settle into dynamic regimes rather than frozen configurations [Gross and Blasius, 2008]. Our criterion detects either an exactly recurring edge set or a plateau in the rate of structural change.

4.4.2 Structural Convergence Metrics

Let $E(t)$ be the directed edge set and $S_i^{(t)}$ agent i ’s out-neighborhood.

Temporal correlation coefficient. Following Tang et al. [2010] and its directed extension [Büttner et al., 2016],

$$C_i^{\text{out}}(t) = \frac{|S_i^{(t-1)} \cap S_i^{(t)}|}{\sqrt{d_i^{\text{out}}(t-1)d_i^{\text{out}}(t)}}, \quad C^{\text{out}}(t) = \frac{1}{N} \sum_i C_i^{\text{out}}(t).$$

Because every out-degree equals K , this is the mean fraction of neighbors retained between consecutive steps.

Normalized Hamming distance.

$$dS(t) = \frac{|E(t) \triangle E(t-1)|}{|E(t)| + |E(t-1)|}.$$

Spectral distance. For the normalized Laplacian $L = I - D^{-1/2}A_u D^{-1/2}$ of the undirected projection [Chung, 1997], with ascending eigenvalue vector $\lambda(t)$,

$$d_L(t) = \|\lambda(t) - \lambda(t-1)\|_2.$$

This distinguishes changes that have different global structural effects even when their edge-level Hamming distances are similar.

DeltaCon. Following Koutra et al. [2013], Fast Belief Propagation forms $F(t) = [I + \epsilon^2 D - \epsilon A]^{-1}$ with $\epsilon = 1/(1 + d_{\max})$. The Matusita distance

$$d(t) = \sqrt{\sum_{ij} \left(\sqrt{f_{ij}^{(t-1)}} - \sqrt{f_{ij}^{(t)}} \right)^2}$$

gives similarity $\text{sim}(t) = 1/(1 + d(t))$. We also record the Adjusted Rand Index between consecutive Louvain partitions [Hubert and Arabie, 1985].

4.4.3 Stopping Rule and Attractor Classification

A run stops at

$$t_{\text{conv}} = \min\{t : \text{exact recurrence or } C^{\text{out}} \text{ plateau}\}.$$

Exact recurrence requires $E(t) = E(t - p)$ to hold persistently for some $p \in [1, P_{\text{max}}]$. The plateau condition requires, for patience consecutive steps,

$$\frac{|\text{mean}(C^{\text{out}}[t - W : t]) - \text{mean}(C^{\text{out}}[t - 2W : t - W])|}{\text{mean}(C^{\text{out}}[t - 2W : t - W])} < \varepsilon_C.$$

After detection, the simulation records `post_window` additional steps; T_{max} is a hard cap.

Runs are classified as `fixed_point` when the edge set becomes stationary, `limit_cycle:p` when it recurs with $2 \leq p \leq P_{\text{max}}$, `plateau` when the retention rate stabilizes without a detected period, and `none` when no condition is met by T_{max} . These labels describe observed graph dynamics. They do not by themselves establish a Nash equilibrium or invoke the OGA guarantees in Section 5, because the simulated agents do not run OGA. A plateau may represent an aperiodic regime or a cycle longer than P_{max} .

Parameter	Value	Role
W	10	averaging-window length
ε_C	0.01	maximum relative change between windows
<code>patience</code>	5	consecutive steps satisfying the condition
<code>post_window</code>	10	steps recorded after detection
T_{max}	120	maximum run length
P_{max}	20	longest tested recurrence period

We use C^{out} as the stopping statistic because it is bounded, degree normalized, and smoother than edge-level Hamming distance at $N = 50$. Its plateau height is reported as a convergence diagnostic rather than an ordered structural outcome.

4.4.4 Cross-Run Comparability

Because run lengths differ, outcome metrics are compared at $t_{\text{conv}} + \text{post_window}$, each run’s own evaluation point. Values at fixed indices $t = 20$ and $t = 35$ are retained as robustness checks. The distribution of t_{conv} is itself analyzed across α , topology, and topic.

5 Equilibrium Analysis

In stochastic coevolutionary opinion formation games, proving convergence to standard equilibrium concepts (such as Nash equilibria) is non-trivial. Because network topologies evolve through randomized rewiring, small shifts in expressed opinions may drastically alter global transition probabilities (Chen et al. [2026a,b]).

By framing the hybrid population of analytical cost-minimizing agents ($\mathcal{N}_C$) and LLM generative agents ($\mathcal{N}_L$) as a multi-player general-sum Markov game, we apply Chen et al. [2026a][Theorem 1] to establish strong convergence and social welfare guarantees under Optimistic Gradient Ascent (OGA) learning dynamics (Chen et al. [2026a,b]).

5.1 The Hybrid Game as a Markov Game

We write the Hybrid Coevolutionary Opinion Game as a multi-player general-sum Markov game

$$\mathcal{M}_{\text{hybrid}} = \langle \mathcal{N}, \mathcal{S}, \mathcal{A}, \mathcal{P}, \{r^i\}_{i \in \mathcal{N}}, \gamma \rangle,$$

whose components are the following.

Players. $\mathcal{N} = \mathcal{N}_C \cup \mathcal{N}_L$, with $|\mathcal{N}_C| = \lfloor \alpha n \rfloor$ Type-C agents and $|\mathcal{N}_L| = n - \lfloor \alpha n \rfloor$ Type-L agents.

States. A global state at step t is the pair $S^{(t)} = (G^{(t)}, \mathbf{X}^{(t-1)})$, where $G^{(t)}$ collects the active out-edges and $\mathbf{X}^{(t-1)}$ the texts of the previous step. The set of finite strings over a vocabulary is countably infinite, so we take the state space to be the interaction graphs alone. Each of the n agents holds a K -subset of the other $n - 1$, which gives

$$|\mathcal{S}| = \binom{n-1}{K}^n < \infty,$$

the finiteness the cited result requires. Texts then enter only through the opinions they are scored to.

Actions. $\mathcal{A} = \prod_{i=1}^n \mathcal{A}_i$. A Type-C agent plays $a_i = z_i \in [-1, +1]$ directly; a Type-L agent plays $a_i = \mathbf{x}_i \in \mathcal{V}^*$, which the stance map carries to $z_i = e(\mathbf{x}_i) \in [-1, +1]$. Both types therefore act on one scale.

Transition kernel. This is the component the conference version of this paper defers to here. Write $\mathbf{z}^{(t+1)}$ for the opinions the joint action $\mathbf{a}^{(t)}$ is scored to, and $\mathbf{s}$ for the fixed intrinsic opinions. Rewiring snapshots those opinions and then lets every agent rank the same snapshot on its own, so the n draws are conditionally independent and

$$\mathcal{P}(S^{(t+1)} | S^{(t)}, \mathbf{a}^{(t)}) = \prod_{i=1}^n \mathbb{P}(S_i^{(t+1)} | \mathbf{z}^{(t+1)}, s_i), \quad (9)$$

with each factor the Gibbs form of Section 3.3. Two properties of Eq. (9) matter for what follows. It does not depend on $S^{(t)}$, because the rule rebuilds every out-neighborhood from the current opinions rather than editing the previous edge set, so the chain is driven by the opinions alone. And in the $\beta \rightarrow \infty$ limit used throughout the experiments each factor collapses onto the minimisers of the total distance,

$$\mathbb{P}(S_i^{(t+1)} = S | \mathbf{z}^{(t+1)}, s_i) \longrightarrow \frac{\mathbf{1}[S \in \mathcal{M}_i]}{|\mathcal{M}_i|}, \quad \mathcal{M}_i = \arg \min_{|S'|=K} \sum_{j \in S'} |z_j^{(t+1)} - s_i|,$$

which is a point mass on the K -nearest-neighbor set when the K -th and $(K+1)$ -th distances differ, and a uniform draw over the tied peers when they do not. The implementation realises that draw with a dedicated seeded generator, so ties do not fall to the lowest agent index.

Rewards. The reward is the negative stage cost, $r^i(s, \mathbf{a}) = -C_i(G, \mathbf{z})$, and since $z_i, s_i \in [-1, +1]$ it is bounded by $|r^i(s, \mathbf{a})| \leq 4(1 + \rho_i)K$.

To apply Theorem 1 to the hybrid Markov coevolutionary opinion game, we verify that the hybrid learning dynamics satisfy all structural assumptions (Chen et al. [2026a]):

- (i) **Norm Pairs and Diameter Bounds:** Because opinions z_i and policy distributions π_i^s reside in compact, bounded domains, the maximum policy space diameter for any state $s \in \mathcal{S}$ is finite:

$$\Omega_i^s \triangleq \sup_{x, x' \in \mathcal{X}_i^s} \|x - x'\| < \infty$$

The dual norm pair $(\|\cdot\|, \|\cdot\|_*)$ satisfies $\|\cdot\| \geq C\|\cdot\|_1$ and $\|\cdot\|_* \leq C^*\|\cdot\|_\infty$ for constants $C, C^* > 0$ (Chen et al. [2026a]).

- (ii) **Smooth Regularizer and OGA Updates:** Each player updates their state-conditioned strategy profile $\pi_{i,t}^s$ using Optimistic Gradient Ascent with G_i -smooth regularizers $\mathcal{R}_i$ (Chen et al. [2026a]):

$$\begin{aligned}\hat{x}_{i,t+1}^s &= \Pi_\Delta \{ \hat{x}_{i,t}^s + \eta u_{i,t}^s \} \\ x_{i,t+1}^s &= \Pi_\Delta \{ \hat{x}_{i,t+1}^s + \eta u_{i,t}^s \}\end{aligned}$$

where $u_{i,t}^s$ is the optimistic gradient estimator approximating $Q_{i,t}^s \pi_{-i,t}^s$ (Chen et al. [2026a]).

- (iii) **Learning Rate Regime (η):** The learning rate is chosen as $\eta \leq O(1/\sqrt{T})$ to offset the positive regret terms generated by time-varying Q -value functions in general-sum games (Chen et al. [2026a]).
- (iv) **Cumulative Regret Condition:** The non-negativity of cumulative regret $\sum_i R_i^T \geq 0$ is preserved due to the continuous concavity/boundedness of state value functions $V_\pi^i(s)$ under quadratic disagreement costs (Chen et al. [2026a]).

Theorem 1 (Equilibrium and PoA Guarantees in Hybrid Opinion Games). *Consider the hybrid Markov coevolutionary opinion game played by Type-C and Type-L agents running OGA dynamics with learning rate $\eta \leq O(1/\sqrt{T})$.*

For any $\epsilon > 0$, after

$$T > \frac{2}{\epsilon^2 |\mathcal{S}|} \sum_{s \in \mathcal{S}} \sum_{i \in \mathcal{N}} \Omega_i^s$$

iterations, the system satisfies at least one of the following two guarantees (an extension of the result in Chen et al. [2026a]):

1. **ϵ -Approximate Hybrid Nash Equilibrium:** *There exists an iteration $t \in [T]$ such that the joint policy profile $\mathbf{z}_t^{s_t} = (\pi_{1,t}^{s_t}, \dots, \pi_{n,t}^{s_t})$ constitutes an $O(\epsilon)$ -approximate Nash Equilibrium for the network state $s_t = G^{(t)}$. Specifically, the Nash gap satisfies:*

$$\text{Nash-gap}(\mathbf{z}_t^{s_t}) \leq \epsilon \left(C^* \max_i \left\| Q_{i,t}^{s_t} \mathbf{z}_{-i,t}^{s_t} \right\|_\infty + \frac{2 \max_i \{G_i \Omega_i^s\}}{\eta} \right)$$

2. **Bounded Price of Anarchy (PoA):** *Alternatively, the empirical average social welfare over time horizon T is strictly bounded relative to the optimal central-planner policy profile π^* , where (λ, μ) are the smoothness constants of the game. We write them λ and μ rather than the (α, β) of the source, since α is the mixing ratio and β the choice intensity throughout this paper (Chen et al. [2026a]):*

$$\frac{1}{T} \sum_{t=1}^T C_{\pi_t}(s) \leq \frac{\lambda}{1 + \mu} C_{\pi^*}(s) - \frac{1}{1 + \mu} \frac{\epsilon^2}{8\eta}$$

6 Experiments and Numerical Results

6.1 Dataset Overview

We source data from the `r/politics` subreddit via PushShift’s [Baumgartner et al., 2020] April 2019 dump, containing 138.5 million comments. After streaming decompression and filtering (non-deleted, non-removed, body length > 20 characters), 122.5 million active comments remain (deletion rate: 11.56%). Semantic clustering (Section 4.2) identifies 1,184 topic clusters across politics; we focus on two politically contested topics relevant to our simulation: *gun control* ($n = 3,581$ comments) and *abortion* ($n = 1,618$ comments).

Figure 2 shows the stance distributions for both topics under the regressor of Section 4.2. Gun control exhibits a near-neutral mean with high variance, consistent with a bimodal and divided public; abortion displays a more concentrated opposing consensus. The resulting initial-stance split among the $N = 50$ sampled agents is 54/46 for gun control and 68/32 for abortion, an asymmetry that Section 6.5 shows to be consequential.

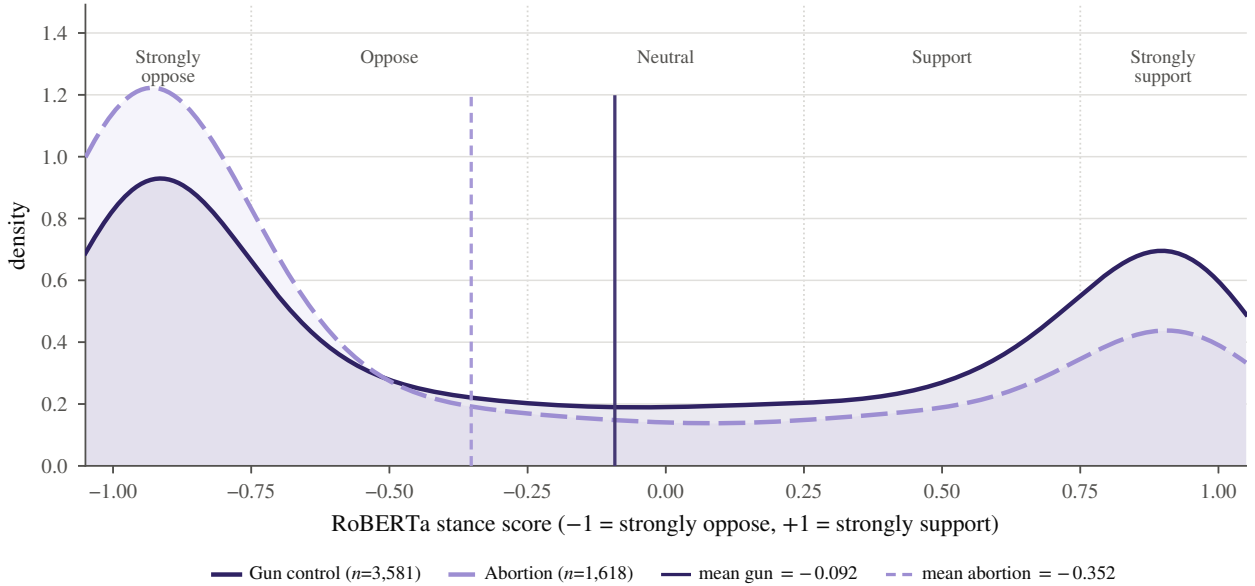


Figure 2: Stance score distributions for gun control ($n = 3,581$) and abortion ($n = 1,618$) comments in `r/politics` (April 2019). Gun control shows a bimodal, polarized distribution, while abortion concentrates toward the opposing end of the spectrum, consistent with the 54/46 and 68/32 initial-stance splits in the sampled agent population.

The two topics are not independent replicates. The 50 sampled agents share identical personas and identical stubbornness coefficients ρ_i across the two topics; only the initial stance differs. Measured across the population, $\text{corr}(s^{\text{gun}}, s^{\text{abortion}}) = +0.881$. Any topic effect is therefore attributable to the stance distribution rather than to a different population of agents, and the two topics are compared with paired rather than independent-sample tests (Section 6.3). Assigning the same individual identical stubbornness on both issues is a modeling simplification, not an empirical finding.

6.2 Simulation Configuration

All simulations use $N = 50$ agents and $K = 5$ neighbors per agent. The mixing parameter is varied across $\alpha \in \{0, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875, 1\}$, covering the full spectrum from a purely LLM-driven population ($\alpha = 0$) to a purely cost-minimizing population ($\alpha = 1$).

Topology parameters. The three initial-topology families of Section 4.3.2 are matched on mean degree (3.84 for BA, 3.84 for ER, 4.00 for WS; maximum deviation 4.17%), so that any difference between them is attributable to structure rather than to edge count. Their clustering coefficients remain clearly distinct at 0.179, 0.079, and 0.390 respectively, preserving the contrast of interest.

Grid completeness. The design yields $n = 60$ runs at every α level and $n = 30$ at every (α, topic) cell, with no missing cells: all 540/540 runs complete and enter the analysis.

Realized simulation length. Runs terminate under the stopping rule of Section 4.4.3. Mean settling time at $\alpha = 0$ is 52.2 steps, and the longest run lasts 115 steps in total, including the post-detection window, within the $T_{\max} = 120$ horizon. Run length is itself a function of composition and is reported in Section 6.5.1.

6.3 Statistical Analysis

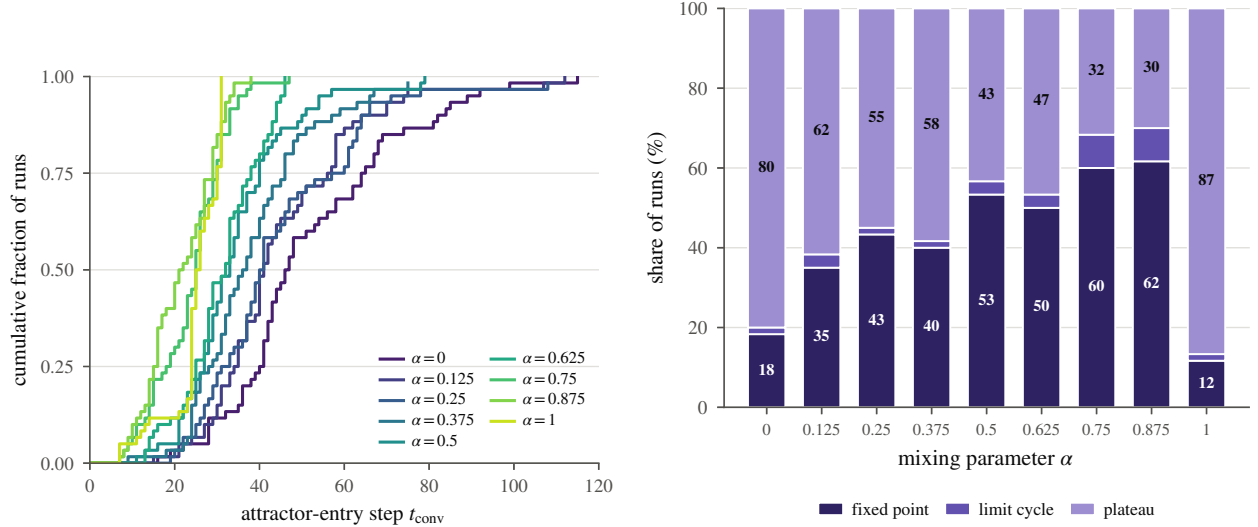
Variances are markedly unequal across α , with the standard deviation of PoA falling by two orders of magnitude between $\alpha = 0$ and $\alpha = 1$, so comparisons between α levels use Welch’s t -test [Welch, 1947] with Benjamini–Hochberg FDR control [Benjamini and Hochberg, 1995] over the family of adjacent- α comparisons, and we report q -values throughout.

Where the claim of interest is equivalence rather than difference, specifically for the comparison among initial topologies, we test for it directly with two one-sided tests [Schuirmann, 1987] against a margin fixed in advance at $\delta = 0.2$ PoA units. The margin is absolute, so its relative size varies across the range: 17.5% of PoA at $\alpha = 1$ but 3.6% at $\alpha = 0$. We fixed it before seeing the data and report the consequence in Section 6.5.5.

Comparisons between topics use paired tests, for the reason given in Section 6.1. All confidence intervals are reported at the 95% level and quantify variability across topologies and Type-C/Type-L assignments rather than across draws of the agent sample (Section 4.3.2).

6.4 Implementation Details

Type-L agents are powered by Phi-4 (14.7B parameters) [Abdin et al., 2024] under 4-bit AWQ quantization [Lin et al., 2024], served locally via vLLM [Kwon et al., 2023] on a single RTX 4090. Type-C agents are implemented as closed-form FJ updates requiring no neural inference. Network operations (K-NN rewiring, Louvain modularity, Laplacian computation) are performed with NetworkX and python-louvain. Ties at the K-NN cutoff are broken by seeded random choice, under which the correlation between in-degree and agent index is -0.037 . The numerical ($\alpha = 1$) path is bit-for-bit reproducible across repeated runs, and an audit of all 540 runs confirmed complete, schema-conformant per-step output with reasoning traces retained in full. The full simulation codebase is available at <https://github.com/Evan-Jiang/Echo-Chamber-Simulation>.



(a) Attractor-entry step t_{conv} , as empirical cumulative distribution functions.

(b) Attractor composition. Exactly stationary edge sets are rare at both pure compositions and most common in the mixed regime.

Figure 3: Settling behavior by mixing ratio ($n = 60$ per curve and per bar, pooled over three topologies and two topics).

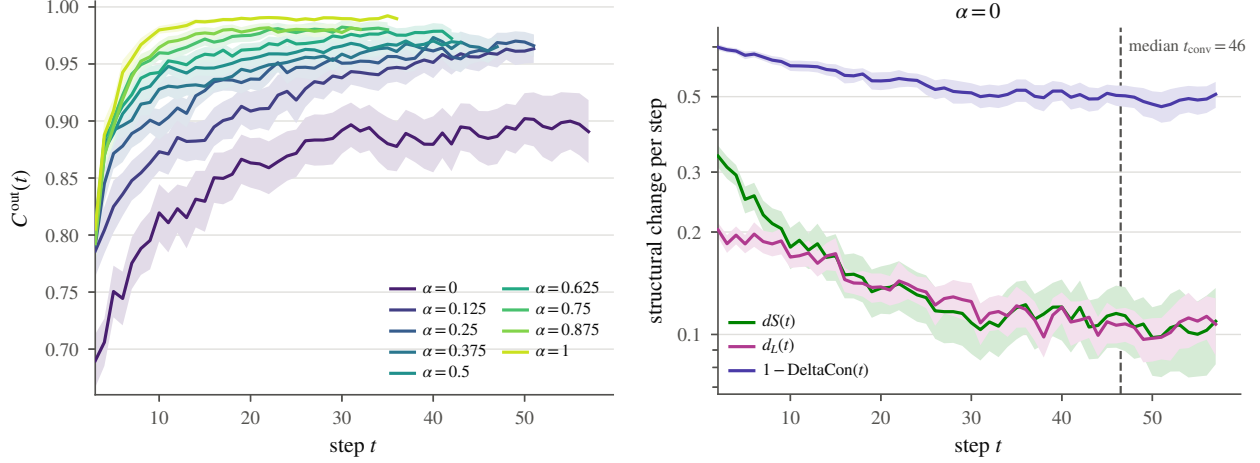
6.5 Numerical Results

6.5.1 Convergence of the interaction structure

Whether the coevolving graph settles is an empirical question for both agent types, and it is not answered by the analytical properties of either. The Friedkin–Johnsen update has a known fixed point, but K-NN rewiring is a discontinuous map applied on top of it, so convergence of the opinion vector does not imply convergence of the interaction structure. For Type-L agents there is no guarantee at either level: positions are obtained by scoring generated text, and nothing in the update rule constrains the sequence of graphs to settle. That both modes of updating converge is therefore a result, and it is what licenses comparing converged states across α at all. Theorem 1 guarantees that one of two regimes is reached; what the grid establishes is which of them, and how often.

Both pure populations converge, and to the same kind of attractor. Figure 3 summarizes settling behavior across the grid. Neither pure population settles predominantly to an exact fixed point: both are classified plateau in the large majority of runs, and the purely Type-L population reaches an exactly stationary edge set slightly more often than the purely Type-C one, whose opinion dynamics converge analytically. Under K-NN rewiring, then, convergence takes the form of a plateau in the rate of structural change rather than a frozen graph, and Type-L agents attain that state at least as reliably as the closed-form Type-C dynamics. The two modes differ in how long they take and in the level at which they settle, examined next, but not in whether they settle or in what they settle into.

Settling time is a non-monotone function of composition. Figure 3a shows t_{conv} falling monotonically in α from 52.2 steps at $\alpha = 0$ to 21.7 at $\alpha = 0.875$, and then rising to 25.1 at $\alpha = 1.0$. A population that is almost entirely cost-minimizing settles faster than one that is entirely cost-minimizing. The dispersion is also composition-dependent: the longest run at $\alpha = 0$ lasts 115 steps in total, post-detection window



(a) Temporal correlation coefficient $C^{\text{out}}(t)$, the fraction of out-neighbors retained between consecutive steps; markers give the median t_{conv} . (b) Three measures at $\alpha = 0$ on a logarithmic vertical axis, all flattening ahead of the median t_{conv} (dashed).

Figure 4: Structural change per step, defined from $t = 2$ (mean with 95% CI band). Panel (a) is the statistic the stopping rule reads; panel (b) shows three measures that take no part in it.

included, against a mean t_{conv} of 52.2, while at $\alpha = 1$ the entire range of t_{conv} spans 7 to 31.

Attractor composition shows the same reversal. Figure 3b resolves the composition by class. Exactly stationary edge sets account for 18% of runs at $\alpha = 0$ and 12% at $\alpha = 1$, whereas mixtures reach 62% at $\alpha = 0.875$: a population containing six LLM agents attains an exact fixed point five times as often as one containing none. We report this without a mechanism; one testable conjecture is that a small number of Type-L agents supplies fixed reference points that halt the continual micro-adjustment of Type-C agents, allowing an exact fixed point to be reached rather than approached.

Rewiring churn separates the purely Type-L population. Figure 4 traces the per-step structural change, and Figure 4a the stopping statistic itself. The plateau height of C^{out} is 0.913 ± 0.069 at $\alpha = 0$, corresponding to roughly 8.7% of out-neighbors replaced per step, against 0.990 ± 0.004 at $\alpha = 1$ and roughly 1%. The interior levels lie flat between 0.971 and 0.983, so elevated churn under Type-L updating is confined to $\alpha = 0$ rather than declining smoothly across the range, and the distinctive behavior at $\alpha = 1$ visible in the other two measurements is a difference in attractor type rather than in churn. The two extremes therefore settle into regimes that differ in kind as well as in degree: the Type-C population comes to rest close to a frozen structure, whereas the Type-L population reaches a state in which the rate of turnover has stabilized while a small fraction of ties continues to change.

Measures outside the stopping rule corroborate it. Convergence is declared on C^{out} , and three quantities that take no part in that decision agree with it. The normalized Hamming distance between consecutive edge sets, the spectral distance between their Laplacians, and $1 - \text{DeltaCon}$ are recorded at every step transition but are never read by the rule. Figure 4b shows all three at $\alpha = 0$, the slowest-settling composition: each falls substantially and then flattens at around $t = 25$ to 30, ahead of the median t_{conv} of 46. Four measures of structural change therefore locate the same settled regime, and the one the rule reads is the last of them to register it. The convergence criterion is satisfied within the simulation horizon in all runs (540/540), and therefore no runs are excluded.

Table 1: Metrics at each run’s converged state ($t_{\text{conv}} + 10$), by topic and mixing ratio. $n = 30$ per row (3 topologies $\times$ 10 seeds), mean $\pm$ 95% CI. $\downarrow$ denotes lower-is-better. Q_{norm} is the null-corrected modularity of Section 3.5 and z_Q the same quantity in units of the null distribution’s standard deviation.

Topic	α	P_z	Q_{norm}	z_Q	PoA $\downarrow$
Gun Control	0	0.261 ± 0.013	0.241 ± 0.009	22.8 ± 2.1	5.681 ± 0.340
	0.125	0.312 ± 0.016	0.250 ± 0.009	23.0 ± 1.5	5.480 ± 0.312
	0.25	0.364 ± 0.013	0.259 ± 0.009	23.8 ± 1.6	5.642 ± 0.396
	0.375	0.409 ± 0.014	0.263 ± 0.009	22.5 ± 1.3	5.477 ± 0.325
	0.5	0.451 ± 0.012	0.277 ± 0.007	23.3 ± 1.3	5.374 ± 0.395
	0.625	0.494 ± 0.011	0.286 ± 0.010	23.1 ± 1.3	4.424 ± 0.344
	0.75	0.538 ± 0.008	0.314 ± 0.008	24.9 ± 1.9	3.821 ± 0.274
	0.875	0.582 ± 0.006	0.350 ± 0.009	24.5 ± 1.3	3.220 ± 0.222
	1	0.619 ± 0.000	0.427 ± 0.002	27.4 ± 1.9	1.149 ± 0.008
Abortion	0	0.347 ± 0.017	0.194 ± 0.006	18.4 ± 1.4	5.436 ± 0.535
	0.125	0.364 ± 0.011	0.199 ± 0.007	20.7 ± 1.6	6.322 ± 0.389
	0.25	0.426 ± 0.015	0.191 ± 0.007	18.8 ± 1.6	6.077 ± 0.367
	0.375	0.463 ± 0.016	0.241 ± 0.013	23.2 ± 1.9	5.815 ± 0.385
	0.5	0.480 ± 0.015	0.283 ± 0.010	24.4 ± 2.3	5.117 ± 0.298
	0.625	0.510 ± 0.013	0.297 ± 0.008	24.0 ± 1.9	4.670 ± 0.296
	0.75	0.534 ± 0.010	0.317 ± 0.008	24.4 ± 1.5	3.853 ± 0.288
	0.875	0.554 ± 0.008	0.343 ± 0.006	25.0 ± 1.4	2.758 ± 0.306
	1	0.575 ± 0.000	0.419 ± 0.002	25.8 ± 1.4	1.129 ± 0.001

Convergence is slower on the near-balanced topic. Gun control settles later than abortion at every α ($t_{\text{conv}} = 58.7$ against 45.7 at $\alpha = 0$). This follows from the initial stance distributions: gun control is near-balanced at 54/46, leaving many agents close to the K-NN decision boundary where small changes in expressed opinion continue to flip edges, whereas abortion has a clear majority at 68/32 and settles sooner. The same asymmetry reappears in the efficiency response of Section 6.5.3.

All metrics below are read at each run’s own converged state rather than at a common step index.

6.5.2 Final-state comparison

Table 1 is the primary result. Polarization and null-corrected modularity rise monotonically in α while PoA falls, and the confidence intervals are narrow throughout. Figure 5 shows the same quantities over time: each settles within roughly ten steps and remains flat thereafter, so the reported values are insensitive to the exact read-out step. The three responses are read in turn below.

6.5.3 Efficiency and the Price of Anarchy

At $\alpha = 1$ the dynamics reduce to the FJ closed form, whose fixed point is the planner’s optimum $z^* = (L_{\text{sym}} + W)^{-1}Ws$; the residual inefficiency is the gap between the Nash outcome and the optimum of the opinion formation game. The measured PoA = 1.129 ± 0.001 (abortion) and 1.149 ± 0.008 (gun control) sit close to the $9/8$ bound of Bhawalkar et al. [2013]. That bound does not strictly apply to a directed graph whose edges are counted once. The agreement is still worth noting: it is an external reference point for the Type-C path.

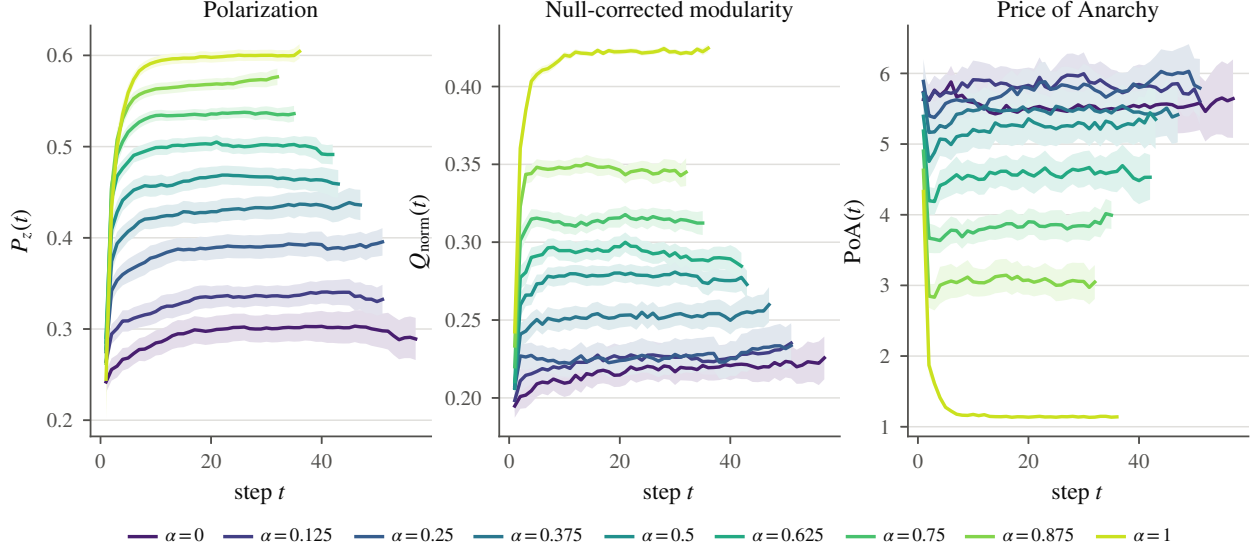


Figure 5: Per-step trajectories of the three reported metrics by mixing ratio (mean with 95% CI band). All three are flat well before the graph enters its attractor, so the values in Table 1 do not depend on the precise step at which they are read.

Pooled over topologies and topics, PoA falls from 5.558 ± 0.309 at $\alpha = 0$ to 1.139 ± 0.005 at $\alpha = 1$. Table 2 shows this is not a smooth interpolation: no comparison up to and including the one reaching $\alpha = 0.375$ is significant ($q > 0.12$), and every comparison from $\alpha = 0.5$ onward is ($q < 0.04$). The low- α region is a statistical plateau, and the decline begins only once cost-minimizing agents approach parity. PoA is minimized at $\alpha = 1$ and no interior mixture approaches it; since polarization is lowest exactly where efficiency is worst, no composition optimizes both.

Table 3 and Figure 6 separate the two components. Between $\alpha = 1$ and $\alpha = 0$ the disagreement term grows by a factor of 3.3 ($0.956 \rightarrow 3.136$) while the conformity term grows by a factor of 13.2 ($0.183 \rightarrow 2.422$). Conformity accounts for 43.6% of social cost in a purely Type-L population against 16.1% in a purely Type-C one, and its share declines monotonically across all nine levels. The characteristic inefficiency of Type-L agents is thus displacement from their own intrinsic position, not disagreement with neighbors.

Within the plateau the two topics differ. Pooled, the rise from $\alpha = 0$ to $\alpha = 0.125$ is not significant ($p = 0.095$); separated, abortion rises from 5.436 to 6.322 ($t = 2.74$, $p = 0.0084$, $d = 0.71$) while gun control is flat and attains its maximum at $\alpha = 0$ ($p = 0.379$). Because the two topics share agent personas and stubbornness coefficients (Section 6.1), the difference is attributable to the stance distribution rather than to population composition.

6.5.4 Echo chamber formation

P_z rises from 0.304 ± 0.015 to 0.597 ± 0.006 and Q_{norm} from 0.218 ± 0.008 to 0.423 ± 0.002 across the range of α . Echo chamber structure forms under every composition, but its strength depends on agent type: Type-L populations produce measurably weaker polarization and weaker community structure. This is the mechanism of the cost decomposition seen structurally: a population readily displaced from its intrinsic positions does not hold camp boundaries, and camps that do not hold do not consolidate.

All claims use Q_{norm} rather than bare Q . At $\alpha = 1$ the bare value is 0.763 while the degree-preserving null attains 0.336 on the same degree sequence, so roughly 44% of raw modularity follows from the constant out-degree the K-NN rule imposes. What remains is far from null at every composition: z_Q places the

Table 2: Price of Anarchy by mixing ratio, pooled ($n = 60$ per row). The q column gives the Welch test against the preceding level under Benjamini–Hochberg control.

α	PoA	q vs. prev.
0.0	5.558 ± 0.309	—
0.125	5.901 ± 0.265	0.127
0.25	5.860 ± 0.268	0.826
0.375	5.646 ± 0.249	0.280
0.5	5.246 ± 0.242	0.036
0.625	4.547 ± 0.222	8.5×10^{-5}
0.75	3.837 ± 0.193	1.1×10^{-5}
0.875	2.989 ± 0.193	3.0×10^{-8}
1.0	1.139 ± 0.005	3.9×10^{-26}

Table 3: The same social cost separated into its two components, pooled ($n = 60$ per row). Their sum is the Price of Anarchy at left.

α	Disagreement	Conformity	Conf. share
0.0	3.136	2.422	43.6%
0.125	3.690	2.211	37.5%
0.25	3.882	1.978	33.7%
0.375	3.822	1.824	32.3%
0.5	3.610	1.636	31.2%
0.625	3.127	1.420	31.2%
0.75	2.678	1.159	30.2%
0.875	2.175	0.814	27.2%
1.0	0.956	0.183	16.1%

Table 4: PoA by initial topology ($n = 20$ per cell), with the largest pairwise gap at each level and its two one-sided test against $\delta = 0.2$.

α	BA	ER	WS	largest gap	TOST p
0.0	5.348 ± 0.503	5.766 ± 0.737	5.562 ± 0.395	−0.418	0.694
0.25	5.944 ± 0.356	5.961 ± 0.531	5.673 ± 0.557	+0.288	0.594
0.5	5.078 ± 0.373	5.422 ± 0.417	5.237 ± 0.521	−0.344	0.704
0.75	3.912 ± 0.420	4.051 ± 0.332	3.549 ± 0.247	+0.501	0.932
1.0	1.138 ± 0.008	1.134 ± 0.007	1.144 ± 0.010	−0.006	< 0.0001

observed modularity between 18 and 27 standard deviations above the null distribution across the grid, rising with α alongside Q_{norm} itself. The community structure is thus unambiguously present throughout, and the correction bears on how strong it is rather than on whether it exists.

6.5.5 Initial topology

Table 4 reports PoA by family. Across all nine α levels and both PoA and P_z , the confidence intervals of the three topology families overlap and no comparison attains significance; at $\alpha = 0$, P_z is 0.299, 0.299, and 0.314 for BA, ER, and WS respectively. Because the claim of interest here is equivalence, we test for it directly against the margin $\delta = 0.2$ fixed in advance (Section 6.3). Equivalence is established at $\alpha = 1$ ($p < 0.0001$) and at no other level: at low α the standard deviation of PoA is on the order of 1.0 to 1.5, and the data cannot exclude a true difference of up to roughly 0.5. Confirming equivalence across the whole range would require a relative margin fixed in advance, or more seeds.

What the data do support is a mechanism claim. The three families are matched on mean degree while their clustering coefficients differ by a factor of five (Section 6.2), and K-NN rewiring overwrites the initial edge set at the first step. That no difference is detectable under this contrast, at $\alpha = 0$ as much as at $\alpha = 1$, indicates that echo chamber formation here requires no pre-existing hub structure and is not a property of the FJ closed form.

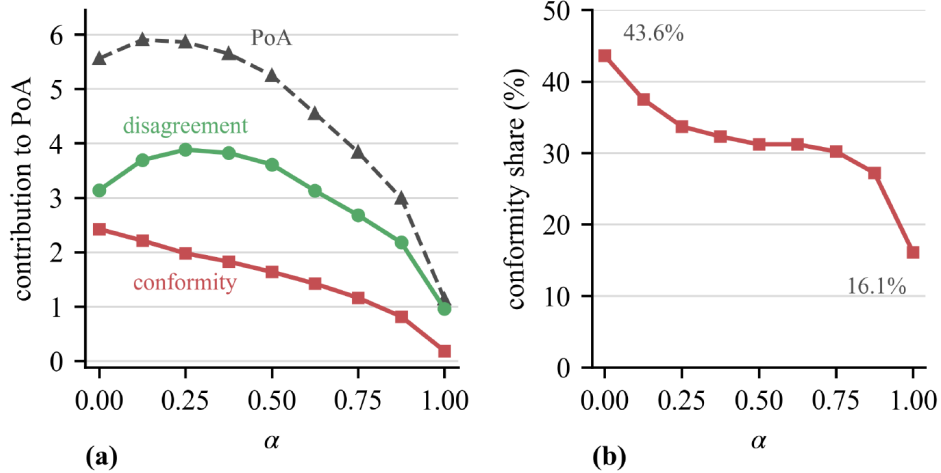


Figure 6: Social-cost decomposition against the mixing ratio, on the values of Table 3. (a) The disagreement term rises to a peak at $\alpha = 0.25$ before falling, while the conformity term declines throughout; each component is divided by the planner optimum of the same step, so the two sum to the Price of Anarchy, shown dashed. (b) The conformity share falls from 43.6% to 16.1%.

7 Discussions, Conclusions and Future Work

7.1 Interpretation of Key Findings

Language reasoning operates as a conformity mechanism. Social cost separates into disagreement with one’s neighbors and displacement from one’s own prior position, and the two behave differently as composition changes. Displacement grows by more than an order of magnitude between the Type-C and Type-L extremes, and becomes the dominant share of the inefficiency in a purely Type-L population. The mechanism is where the anchor sits. In the Type-C cost function an agent’s prior position is a term in the objective, a restoring force that penalizes departure however uniform its neighborhood becomes. A Type-L agent receives the same prior as a sentence in a prompt, and within the homogeneous neighborhoods that rewiring produces, reasoning over what its neighbors say moves it away from that prior rather than defending it. Under uniform peer pressure, language reasoning operates less like deliberation and more like repeated persuasion.

Weaker polarization is a symptom, not an improvement. The structural result follows from the same property. A population that is readily talked out of its own positions does not hold camp boundaries, and camps that do not hold do not consolidate. Weaker polarization among Type-L agents is therefore not evidence of better deliberation; it is a by-product of what makes them inefficient. This is worth stating plainly, because polarization is the metric this literature reports, and read alone it points the wrong way: the composition that looks healthiest by that measure is the one furthest from what a planner could achieve, and the composition closest to the planner’s allocation is the most segregated one. Low polarization is not a free good. It is what a population looks like when its members do not hold their ground.

Two features do not follow this pattern. Mixed populations settle into exactly stationary structures more often than either pure one. And adding a small minority of Type-C agents raises social cost on one of our two topics and not the other; since both topics share personas and stubbornness coefficients, this belongs to the shape of the stance distribution rather than to who the agents are. Both are findings in need of a

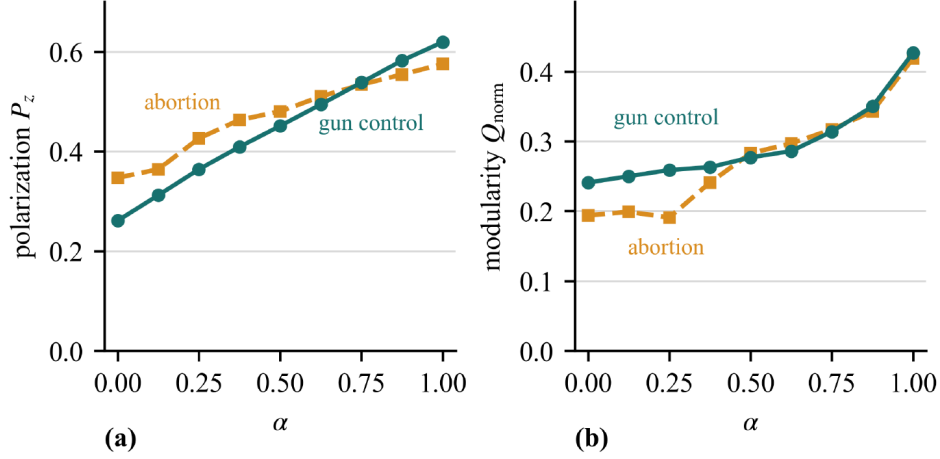


Figure 7: Polarization and echo-chamber structure against the mixing ratio, by topic. (a) The variance of expressed opinions rises with α for both topics. (b) The null-corrected modularity rises over the same range. Both measures move opposite to the Price of Anarchy of Fig. 6, which is why the composition that looks healthiest on polarization is the one furthest from the planner optimum. Both vertical axes start at zero.

mechanism, and we report them as such.

7.2 Conclusions

This work began from a tension between two literatures. Game-theoretic models of opinion formation buy their analytical results with a convex cost function and an update rule chosen to be tractable; agent simulations buy semantic realism by giving that up. The tension is usually described as one between rigor and realism, but the substantive question underneath is narrower: whether equilibrium analysis survives the substitution. A Price of Anarchy is a statement about equilibria, and it says nothing about a population that does not reach one.

What we find is that it does survive. Theorem 1 guarantees that the hybrid dynamics reach either an approximate equilibrium or a regime of bounded long-run average cost, and the grid shows which: the coevolving graph enters an attractor under Type-L updating as reliably as under an update rule with a known fixed point, even though nothing in the construction guarantees it: the rewiring map is discontinuous, and an agent whose position is read off generated text has no closed form to converge to. Opinion dynamics driven by language are, in this respect, well-behaved. That is what allows the efficiency gap we report to be read as a difference between two equilibria rather than between two snapshots, and what allows an agent that reasons in sentences to be placed in the same analytical frame as one that minimizes a quadratic cost.

7.3 Limitations and Future Work

1. All Type-L agents here are instances of a single model. Whether the efficiency gap is a property of language-based opinion updating in general, or of this model’s particular disposition to be persuaded, is not settled by our design. The decomposition above is the instrument that would settle it, and applying it across models is the most direct extension of this work.
2. The population is a stratified sample of $N = 50$ agents whose intrinsic opinions, personas, and stubbornness coefficients are fixed within each topic. Our confidence intervals therefore quantify variability across topologies and agent-type assignments but not across draws of the sample, and the two topics, which share personas by construction, are not independent replicates.

3. Statistical equivalence among the three initial topologies is established only for the purely Type-C population. Elsewhere no significant difference is found, but run-to-run variability at low mixing ratios exceeds what the pre-registered margin can resolve; confirming equivalence across the range would require a relative margin or more seeds.
4. The community structure is subject to the resolution limit of modularity maximization at this network size, which our null model does not correct. The bias runs toward merging communities that should remain distinct, so the reported values are conservative with respect to the compartmentalization they demonstrate.

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